

Team Number: 4

Project Title:

Syria WheatWatch: Drought Impact on Cereal Production

Climate Challenge: Agriculture and food systems

Problem Statement:

Syria experienced its worst drought in nearly 36 years during the 2024-2025 season, with rainfall more than 50% below the long-term average. The result was catastrophic: wheat production fell to 900,000 tonnes—the lowest in over half a century—leaving a 3 million tonne wheat deficit and pushing millions into food insecurity. This project will analyze the relationship between rainfall deficits and cereal production losses in Syria from 1991 to 2025, identifying the rainfall thresholds below which crop failure becomes imminent. Using Python to process climate and agricultural data, the analysis will produce visualizations showing how precipitation variability has driven production declines and quantify the severity of the current crisis. Findings will inform a proposed early-warning framework to help agricultural planners anticipate drought-driven food shortfalls.

Main Research Question:

How has rainfall variability between 1991 and 2025 affected wheat and barley production in Syria, and what precipitation thresholds predict crop failure?

Dataset:

NASA POWER (Precipitation) + FAOSTAT (Crop Production) + World Bank Climate Change Knowledge Portal (Historical Climate)

Source/Link:

- NASA POWER: <https://power.larc.nasa.gov/>
- FAOSTAT Syria Crops: <https://data.humdata.org/organization/fao>
- World Bank CCKP: <https://climateknowledgeportal.worldbank.org/country/syrian-arab-republic>